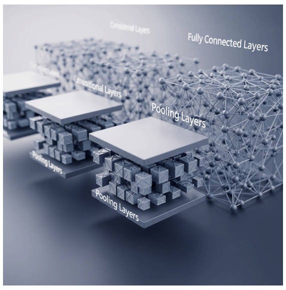
A4.3.9 Convolutional Neural Networks (CNNs)



A4.3.9 Describe how CNNs are designed to adaptively learn spatial hierarchies of features in images

A4.3.9 describes how Convolutional Neural Networks (CNNs) adaptively learn spatial hierarchies of features in images. Their basic architecture includes input, convolutional, activation, pooling, fully connected, and output layers. The number of layers, kernel size, stride, activation function, and loss function all affect how CNNs process data and classify images

Introduction

We've learned about Artificial Neural Networks (ANNs) and how they are inspired by the brain. But a standard Multi-Layer Perceptron (MLP) has a major weakness: it struggles with images. To understand an image, a computer needs to understand spatial relationships, and that's where a special, powerful architecture comes in: the **Convolutional Neural Network (CNN)**.

Why MLPs Fail at Images

An MLP requires a flat, 1D vector of inputs. To feed it an image (a 2D grid of pixels), you must "flatten" it—unravelling the grid into one long line of pixels.

- **The Problem:** This process completely destroys the **spatial hierarchy** of the image. The pixels that form an image are no longer "next to" each other in the data. The model loses all information about the structure of the image, making it very difficult to learn to recognise objects.

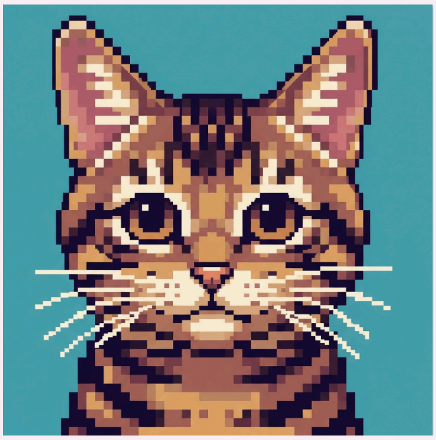
How CNNs Learn: The Spatial Hierarchy of Features

CNNs are designed to solve this problem by mimicking how the human visual cortex works. They don't look at pixels individually; they look at groups of pixels to learn patterns, and then build more complex patterns from simpler ones.

This is the **spatial hierarchy of features**:

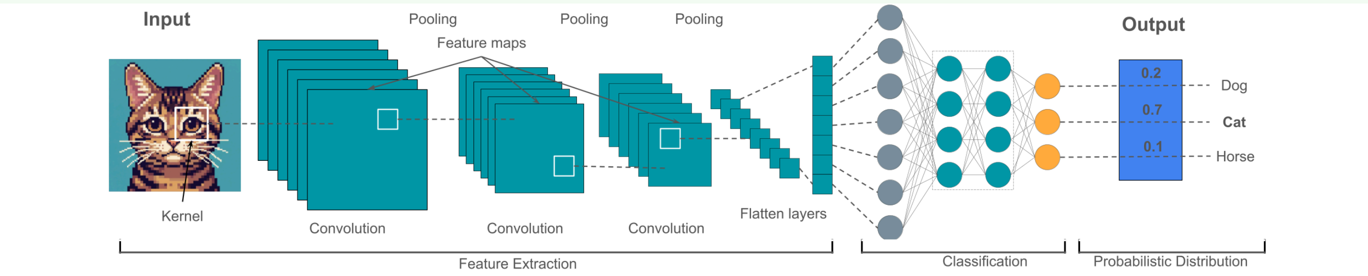
- **Low-Level Features:** The first layers of the network learn to recognise very simple patterns, like diagonal edges, vertical edges, colours, and simple curves.

- **Mid-Level Features:** The next layers combine these simple features to learn more complex shapes, like corners, circles, or textures.
- **High-Level Features:** Deeper layers combine these shapes to learn to recognise parts of objects, like an eye, a nose, or a wheel.
- **Final Classification:** The final layers take all this evidence and make a classification: "The combination of these features strongly suggests this image contains a 'Cat'."



The Basic Architecture of a CNN

A CNN is a multi-stage pipeline. An image goes in one end, and a classification comes out the other. It's made up of several specialised layers.



The CNN Architecture

1. Input Layer

This is simply the raw pixel data of the input image (e.g., a 28x28 pixel grid).

2. Convolutional Layers

This is the core building block of the CNN. Its job is to detect features.

- **Kernel (or Filter):** A small matrix of weights (e.g., 3x3) that acts like a "feature detector." You can have a kernel designed to detect vertical edges, another for horizontal edges, and so on. The network *learns* the optimal values for these kernels during training.
- **Convolution:** The kernel slides over every part of the input image, performing a mathematical operation at each step.

- **Feature Map:** The output is a "feature map," which is essentially a new grid showing where the kernel detected its specific feature in the image.

3. Activation Functions

Just like in a standard ANN, an activation function (like ReLU) is applied after the convolution. Its purpose is to introduce non-linearity into the model, allowing it to learn more complex patterns than simple linear relationships.

4. Pooling Layers

The feature maps from the convolutional layer can be very large. A pooling layer's job is to **down-sample** them, making the network more efficient.

- **Max Pooling:** The most common type. It slides a window (e.g., 2x2) over the feature map and, for each patch, keeps only the *maximum* value.
- **Benefits:** This reduces the size of the data, which speeds up computation. It also makes the model more robust to the exact position of a feature in the image (a concept called translational invariance).

5. Fully Connected Layers

After several convolutional and pooling layers have extracted high-level features from the image, the final feature map is flattened into a 1D vector. This vector is then fed into one or more **fully connected layers**, which are identical to the layers in a standard MLP. Their job is to take the high-level feature evidence and perform the final classification.

6. Output Layer

This is the final layer that produces the prediction. For a classification task, it will typically have one neuron for each possible class and use a function (like softmax) to output a probability for each class.

The Effect of Key Parameters

The behaviour of a CNN is heavily influenced by several key parameters:

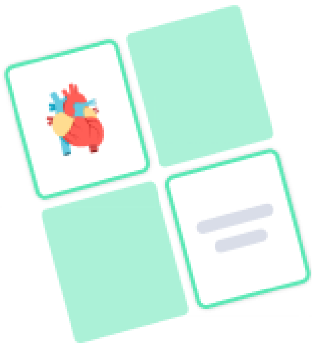
- **Number of Layers:** A "deeper" network (with more convolutional/pooling layers) can learn a more complex spatial hierarchy of features. A shallow network might only learn to detect edges, while a very deep network can learn to recognise specific faces.
- **Kernel Size:** This is the dimension of the filter (e.g., 3x3, 5x5). A smaller kernel will detect finer, more local features, while a larger kernel will detect broader, more general patterns.
- **Stride:** This is the number of pixels the kernel moves at each step. A larger stride (e.g., 2) will result in a smaller feature map and faster computation, but at the cost of less detailed feature extraction.
- **Activation Function Selection:** The choice of activation function (e.g., ReLU, Sigmoid) affects how the network learns and the speed of training. ReLU is the most common choice in modern CNNs.
- **Loss Function:** This is the function that measures how "wrong" the model's prediction is compared to the true label during training. The goal of training is to adjust the model's weights to minimise the value of the loss function. The choice of loss function (e.g., Cross-Entropy for classification) is critical for effective learning.

Flashcards

Use this Quizlet to learn all the key terms in this section.

Match

A 4.3.9 Convolutional Neural Nets



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

View this flashcard set

Choose a Study Mode

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

What is the primary advantage of a CNN over a standard MLP for image recognition?

A. It requires less data to train.

B. It can only be used for classification, not regression.

C. It preserves the spatial hierarchy of features in an image

D. It is computationally faster.

2

In a CNN, what is the name of the small matrix of weights that slides over an image to detect features?

A. A Feature Map

B. A Stride

C. A Pooling Layer

D. A Kernel (or Filter)

3

The primary purpose of a Pooling Layer is to:

A. Down-sample the feature maps to reduce computational complexity.

B. Make the final classification decision.

C. Introduce non-linearity into the model.

D. Detect complex features like eyes or wheels.

4

The concept that early layers in a CNN learn simple features (like edges) and deeper layers learn complex features (like shapes) is called:

A. The activation function

B. The spatial hierarchy of features

C. The loss function

D. Overfitting

5

In a convolutional layer, the parameter that controls how many pixels the kernel moves at each step is the:

A. Kernel Size

B. Loss Function

C. Stride

D. Pooling Size

6

The output of a convolutional layer is a:

A. Final classification probability.

B. 1D vector.

C. Kernel.

D. Feature map.

7

Which layer in a CNN is most like the layers in a standard MLP and is responsible for making the final classification decision?

A. The Input Layer

B. The Convolutional Layer

C. The Pooling Layer

D. The Fully Connected Layer

8

A data scientist is designing a CNN. If they choose a larger kernel size (e.g., 7x7 instead of 3x3), what effect will this likely have?

A. The model will train faster.

B. The model will be less prone to overfitting.

C. The model will only be able to detect fine-grained details.

D. The model will learn to detect larger, more general patterns in the early layers.

9

The function that measures how incorrect a model's prediction is during training is the:

A. Loss Function

B. Pooling Function

C. Kernel Function

D. Activation Function

10

"Max Pooling" is a technique where a window slides over a feature map and:

A. Calculates the average value in the window.

B. Adds a bias term to the window.

C. Multiplies all the values in the window.

D. Keeps only the maximum value from the window.

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